



Format No P013/MT/02

TEST REPORT

Rev. No.01

TEST REPORT ISSUED BY: **BHUBANESWAR LAB**

TEST REPORT NO.

TEST REPORT & ISSUE DATE:

DISCIPLINE : MECHANICAL TEST

CUSTOMER NAME & ADDRESS:

Mr. Sushant Pattnaik, Monoatom Labs

SAMPLE TESTED BY/AT: **BHUBANESWAR LAB**

BNR-1101-TR-823120

01/11/2025

GROUP NAME : BUILDING MATERIALS

PROJECT SITE ADDRESS:

Mr. Sushant Pattnaik, Monoatom Labs

QUANTITY:**DATE OF RECEIPT:****CUSTOMER REF. NO & DATE:****CONDITION OF SAMPLE ON RECEIPT:****SOURCE OF SAMPLE AS GIVEN BY CUSTOMER:****TEST METHOD:**

1 No(s)

14/10/2025

Order No.

Ok

Concrete Mix Design

IS 10262 : 2019, IS 456-2000 RA 2021 Amd-5, IS 9013 : 1978 RA 2013

1. STIPULATION FOR PROPORTIONING

GRADE DESIGNATION: M-30	TYPE OF CONCRETE: RCC
TYPE OF CEMENT : OPC-53	
MAXIMUM NOMINAL SIZE OF AGGREGATE: 20 mm	EXPOSURE CONDITION: MODERATE
WORKABILITY(SLUMP): 150 mm	METHOD OF CONCRETE PLACING: PUMPABLE
TYPE OF AGGREGATE: ANGULAR	BRAND & PRODUCT NAME OF ADMIXTURE: MONOATOM
RETENTION TIME (minutes): 60	TYPE OF ADMIXTURE: PC
CASTING DATE: 31/10/2025	

2. TEST DATA:**DATA OF INGREDIENTS**

Ingredients	Type of material	Source of materials	Average Specific Gravity	Water Absorption, %	DLBD, kg/l
Coarse Aggregate	20 mm		2.89 (ssd condition)	0.16	1.54
	10 mm		2.89 (ssd condition)	0.38	1.54
Fine aggregate	Natural Sand		2.62 (ssd condition)	0.73	1.54
Cement	OPC-53	Ultratech	3.12	-	-
Admixture	PC	Graphene Concrete Admixture	1.08	-	-

RAW MATERIAL ANALYSIS USE FOR CONCRETE MIX

TYPE OF AGGREGATE : Natural Sand



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IS SIEVE DESIGNATION (mm)	Percentage Retain Weight	Percentage Cumulative Retain Weight	Percentage Cumulative Passing Weight	Requirement of % Passing As Per IS 383 , Table No-9 for Fine Aggregate
10	0	0	100	100
4.75	0.7	0.7	99.3	90-100
2.36	2.7	3.4	96.6	75-100
1.18	11	14.4	85.6	55-90
0.6	31	45.4	54.6	35-59
0.3	33.5	78.9	21.1	8-30
0.15	19.2	98.1	1.9	0-10
CONFIRMS TO ZONE : II				

TYPE OF AGGREGATE : 20 mm

IS SIEVE DESIGNATION (mm)	Percentage Retain Weight	Percentage Cumulative Retain Weight	Percentage Cumulative Passing Weight	Requirement of % Passing As Per IS 383 , Table No-7 for Single Size Agg.
40	0	0	100	100
20	15.1	15.1	84.9	85 - 100
10	83.9	99	1	0 - 20
4.75	0.5	99.5	0.5	0 - 5

TYPE OF AGGREGATE : 10 mm

IS SIEVE DESIGNATION (mm)	Percentage Retain Weight	Percentage Cumulative Retain Weight	Percentage Cumulative Passing Weight	Requirement of % Passing As Per IS 383 , Table No-7 for Single Size Agg.
12.5	13.6	13.6	86.4	100
10	26.6	40.2	59.8	85 - 100
4.75	48.4	88.6	11.4	0 - 20
2.36	8.5	97.1	2.9	0 - 5

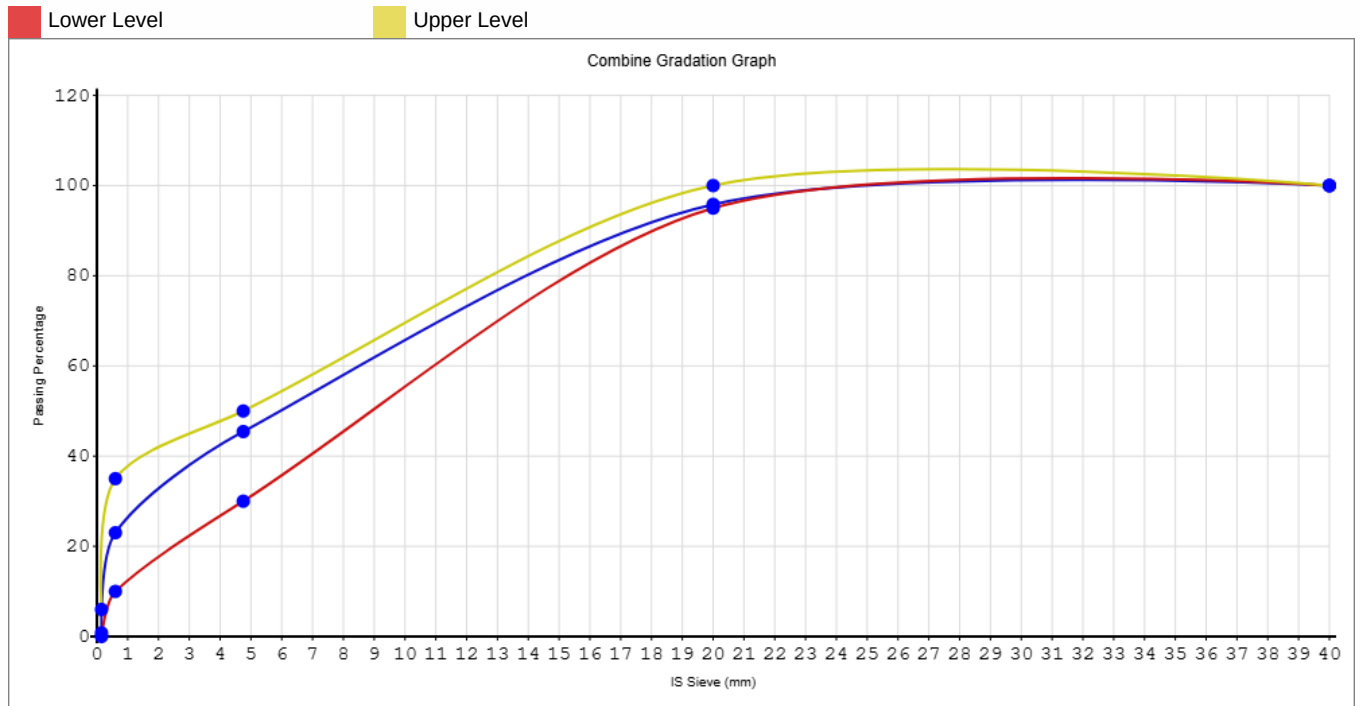
ALL IN AGGREGATE GRADING

IS Sieve DESIGNATION	Percentage passing and proportion of aggregates			As per table 10 of IS 383, percentage passing for All in aggregate grading (20 mm nominal size)
	20 mm	10 mm	River Sand	
	27.8%	30.1%	42.1%	
40	100			100
20	95.8			95 - 100
4.75	45.4			30 - 50
0.6	23			10 - 35
0.15	0.8			0 - 6

ALL IN AGGREGATE GRAPH

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ALL IN AGGREGATE GRAPH



3. TARGET MEAN STRENGTH

a) $f_{ck}' = f_{ck} + 1.65S = 38.25 \text{ N/mm}^2$

where $f_{ck} = 30 \text{ N/mm}^2$ and Standard deviation as per Table no. 2 clause no. 4.2.1.3 of IS 10262 for the grade of concrete M-30 (S) = 5.

b) $f_{ck}' = f_{ck} + X = 36.5 \text{ N/mm}^2$

where $f_{ck} = 30 \text{ N/mm}^2$ and value of X as per Table no. 1 clause no. 4.2.1.3 of IS 10262 for the grade of concrete M-30 (X) = 6.5

The higher value is to be adopted. Therefore, target strength will be 38.25 N/mm^2 is higher than 36.5 N/mm^2

4. AIR CONTENT: As Per Clause No 5.2 Table No. 3 of IS 10262 for Max Aggregate Size 20 mm & grade of concrete M-30 is 1%.

Volume of air content = 0.01 cum

5. SELECTION OF WATER CEMENT RATIO (based on trial experience & also referred figure-1 of IS 10262:2019 & table-5 of IS 456:2000):

Sr No	Adopted Water Cement Ratio (W/c)	Revised W/c
1	0.37	0.37

6. ESTIMATION OF WATER CONTENT: As Per Clause No 5.3 Table No 4 of IS 10262:

6.1: For 50 mm slump using maximum aggregate size 20 mm and texture of aggregate Angular = 186 kg

6.2: Addition of water for workability requirement @ 3% for each 25mm workability above 50 mm workability = 22.3 kg

6.3: Subtraction of water using PC base of plasticizer @ 25% = 52.1 kg.

6.4: Water content (Free Water) = 156.225 kg

7. CALCULATION OF CEMENT CONTENT:

7.1: Adopted Water Cement Ratio (W/c) = 0.37

7.2 Cement Content = 422 kg/m³

7.2a: Revised Water-Cement Ratio = 0.37

7.3: Volume of cement used = 0.135 cum

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8. ESTIMATION OF COARSE AGGREGATE AND FINE AGGREGATE CONTENT:

8.1: Calculation of Volume of Total Aggregate= 0.696 cum.

8.2: Estimation of Coarse Aggregate Content:

8.2.a: For w/c ratio 0.5 and Max Aggregate Size 20 mm and fine aggregate used of zone 2 as per Table No 5 of IS 10262 = $0.62 \times \text{Volume of total aggregate}$.

8.2.b: Addition of coarse aggregate as per clause no 5.5.1 of IS 10262 for revised water cement Ratio (0.37020142180095)= $0.026 \times \text{Volume of total aggregate}$.

8.2.c: Reduction of coarse aggregate as per clause no 5.5.2 of IS 10262 for pumpable mix @ 8% = $0.052 \times \text{Volume of total aggregate}$.

8.2.d: Reduction of Coarse Aggregate for congested reinforcing steel to get desired workability as per clause no 5.5.2 of IS 10262 = 0.04.

8.2.d.1: Volume of coarse aggregate per unit volume of Total Aggregate = 0.554.

8.2.e: Final Volume of coarse aggregate per unit volume of concrete = 0.386 cum.

8.2.1: Volume of 20 mm coarse aggregate used 48% volume of total coarse aggregate = 0.185 cum.

20 mm coarse aggregate content = 535 kg.

8.2.2: Volume of 10 mm coarse aggregate used 52% volume of total coarse aggregate = 0.201 cum.

10 mm coarse aggregate content = 579 kg.

8.3: Calculation for Volume of Fine Aggregate Content: 0.31 cum.

8.3.1.a: Volume of Natural sand used 44.50 % volume of total aggregate: 0.310 cum.

8.3.1.b: Natural sand aggregate content: 812.00 kg.

9.0 Mix Calculation:

a) Total Volume of concrete: 1 cum.

b) Volume of Entrapped Air: 0.01 cum.

c) Volume of Cement: 0.135 cum.

d) Volume of Water: 0.156 cum.

e) Volume of Chemical Admixture: 0.003 cum.

f) Volume of Total Aggregate: 0.696 cum.

g) Mass of Coarse Aggregate: 1,115 kg.

h) Mass of Fine Aggregate : 812 kg.

9.1 CONCRETE MIX PROPORTION USING SSD CONDITION AGGREGATE

Sr No	Ingredients	Material in kg/cum	Percentage proportion of aggregates in SSD condition	Material in kg/Cement bag	Material in Vol (L) / Cement bag	Aggregate by Farma of 35L Cap/ Cement Bag
1	Cement OPC-53	422	-	-		
2	SSD 20 mm	535	27.8	63	41	1.2
3	SSD 10 mm	579	30.1	69	45	1.3
4	SSD River Sand	812	42.1	96	62	1.8
5	Free Water	156	-	19	18.5	
6	Admixture	3.4	0.8% by weight of cement / cementitious content	0.4	0.37	

NOTE - Aggregates shall be used in saturated surface dry condition. If otherwise, when computing the requirement of mixing water, allowance shall be made for the free (surface) moisture contributed by the fine and coarse aggregates. On the other hand, if the aggregates are dry, the amount of mixing water shall be increased by an amount equal to the moisture likely to be absorbed by the aggregates. Necessary adjustments are also required to be made in mass of aggregates.

10. CONCRETE MIX PROPORTION USING DRY CONDITION AGGREGATE

Sr No	Ingredients	Material in kg/cum	Percentage proportion of aggregates in DRY condition	Material in kg/Cement bag	Material in Vol (L) / Cement bag	Aggregate by Farma of 35L Cap/ Cement Bag
1	Cement OPC-53	422	-	-		



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2	DRY 20 mm	535	27.9	63	41	1.2
3	DRY 10 mm	577	30.1	68	44	1.3
4	DRY River Sand	806	42	95	62	1.8
5	Total Water	165	-	20	19.6	
6	Admixture	3.4	0.8% by weight of cement / cementitious content	0.4	0.37	

CONCRETE MIX DESIGN CUBE COMPRESSION TEST RESULTS (ACT BY BOILING WATER METHOD AS PER IS 9013:1978)

CUBE SIZE(mm): 150X150X150

DATE OF CASTING: 31-10-2025

AGE OF SPECIMEN(DAYS):1

DATE OF TESTING : 01-11-2025

Sr. No	Dimension of specimen, mm			Cross sectional area mm ²	Weight, kg	Accelerated Crushing Load, kN	Accelerated Compressive Strength MPa, (Ra)	Equi. 28 Days Comp., (R28=8.09 + 1.64 Ra) Strength, MPa	Type of Failure occurred as per IS 516 Part-1 Sec-1
	Length	Width	Thickness						
1	150.7	150.8	150.4	22724	8.739	551.0	24	47.5	Usual
2	151.1	150.8	151.1	22790	8.692	587.4	26	50.5	Usual
3	150.7	150.4	150.4	22665	8.715	589.2	26	50.5	Usual
Average Value =							25.5	49.5	

Test Witnessed: Mr. Sushant Pattnaik (Scientist).

Remarks: A concrete mix was designed specifically for R&D, adhering to the customer's requirement of including chemical admixture. Workability : Initial Slump-220 mm.

NOTE:

- Mix will require necessary moisture correction of water at site depending on the condition of aggregate (see clause No. 10.2.5 of IS 456:2016)
- As per Clause No.10.2 of IS 456: 2016, Water shall be weighed or measured by volume in a calibrated tank for batching of concrete (see also IS 4925)
- Mix will require necessary bulkgage correction for fine aggregate if volumetric batching.
- As per Clause No.9.2.3 of IS 456: 2016, Mix design done earlier not prior to one year may be considered adequate for later work provided there is no change in source and the quality of the materials.
- Dosage of Admixture is varied (+/- 0.2%) as per the working conditions, environmental conditions (temperature & humidity) and desired workability within the manufacturer recommendations. Performance of admixture shall be checked periodically at site
- The bulk density and specific gravity of materials to be used shall be confirmed by testing of such materials from time to time. If any deviations in the above parameters are observed, suitable corrections for proportions shall be applied at site.
- The coarse and fine aggregate used shall conform to the requirements given in IS 383:2016. The gradation shall be same as that used in the design mix.
- This Certificate valid only to the sample submitted for testing
- Any Correction or changes invalid this Certificate

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H. JOSHI
(Quality Manager)

Authorized Signatory



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